Alternative Rotations

I 3P Rotations

$R(\gamma, \beta, \alpha) = R_x(\alpha) R_y(\beta) R_x(\gamma)$ will rotate through x, y, z in that order

A) Quaterions

- Any rotation can be defined as a rotation about a single axis.
- 3 variables: axis
- 1 variable: rotation

$H = a + bi + cj + dk$ $a, b, c, d \rightarrow$ Real

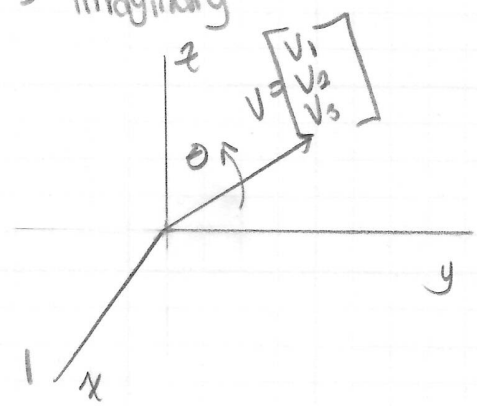
$i, j, k \rightarrow$ imaginary

1) Properties

$ij = k$, $jk = i$ $ki = j$

$i^2 + j^2 + k^2 = -1$

unit vector $\rightarrow \sqrt{a^2 + b^2 + c^2 + d^2} = 1$



$$H = \cos\frac{\theta}{2} + \left(V_1 \sin\frac{\theta}{2}\right) i + \left(V_2 \sin\frac{\theta}{2}\right) j + \left(V_3 \sin\frac{\theta}{2}\right) k$$

↳ Almost unique. It is not because we could do V or $-V$ => 2 unit quanterions represent the same rotation